AI Overview

The **VTK file format** is a standard data format developed by Kitware for the Visualization Toolkit (VTK) to store and share 3D computer graphics, scientific simulation data, and volumetric models. It is natively read and written by visualization tools like ParaView.

Main Format Types

VTK files are divided into three primary format styles: Visualization Toolkit (VTK) +1

1. **Legacy (.vtk)**
   - A serial, plain-text or binary format that is simple to read and write by hand or via custom scripts.
   - Consists of five distinct parts:
     - **Version Identifier**: e.g., `# vtk DataFile Version 3.0`
     - **Header**: Up to 256 characters of descriptive text
     - **File Format**: `ASCII` or `BINARY`
     - **Dataset Structure**: Defines the geometry and topology
     - **Data Attributes**: Point data or cell data (scalars, vectors, tensors) Princeton University +6
2. **XML-Based Formats**
   - More flexible and modern structures supporting compression, random access, and parallel I/O.
   - **Serial extensions**:
     - `.vti` (ImageData / Structured Points)
     - `.vtp` (PolyData / Polygonal Data)
     - `.vtr` (Rectilinear Grid)
     - `.vts` (Structured Grid)
     - `.vtu` (Unstructured Grid)
   - **Parallel extensions**: `.pvti`, `.pvtp`, `.pvtr`, `.pvts`, `.pvtu` (split across multiple files for HPC processing). Visualization Toolkit (VTK) +3
3. **VTKHDF (.vtkhdf)**
   - A high-performance format using HDF5 storage concepts.
   - Built for fast parallel I/O and large-scale scientific datasets. Visualization Toolkit (VTK) +1

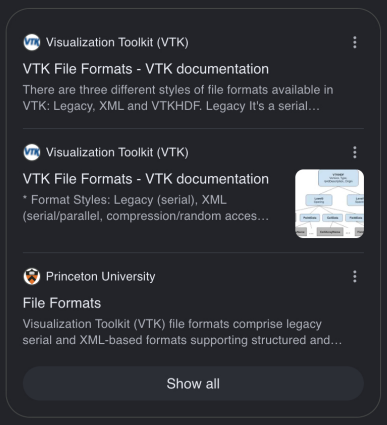
Supported Dataset Types

- **Structured Points / Image Data**: Regular grid of points and voxels.
- **Structured Grid**: Curvilinear grid with implicit topology but explicit coordinates.
- **Rectilinear Grid**: Rectangular grid where coordinates are defined along axis-aligned vectors.
- **Polygonal Data (PolyData)**: Primitives like points, lines, polygons, and triangle strips.
- **Unstructured Grid**: Arbitrary combinations of any 3D cell types (tetrahedra, hexahedra, etc.), common in CFD and finite element analysis (FEA).
- **Field Data**: Multidimensional data not tied to specific geometry or topology. reference.wolfram.com +4

Common Tools & Libraries

- **Visualization / GUI**: ParaView and VisIt.
- **Python / Programming**: Native `vtk` Python bindings, `meshio`, or direct file parsing.
- *Note:* A completely unrelated format also uses the `.vtk` extension—high-frequency 4Hz GPS log files generated by Velocitek marine speed pucks. Wikipedia +5

Are you trying to **read, write, or convert** a VTK file for a specific software application (like ParaView, Python, or OpenFOAM)? Tell me your goal so I can provide the exact syntax or code snippet.



Ask anything

Visualization Toolkit (VTK)
https://docs.vtk.org › vtk_file_formats

VTK File Formats

There are three different styles of file formats available in VTK: **Legacy, XML and VTKHDF**. Legacy#. It's a serial formats that are easy to read and ... Read more

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VTK File Formats

The Visualization Toolkit supports five different dataset formats: **structured points, structured grid, rectilinear grid, unstructured grid, and polygonal data**. Read more

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VTK File Formats

The Visualization Toolkit provides a number of source and writer objects to **read and write popular data file formats**. The. Read more
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2. The VTK file format - VisIt User Manual - Read the Docs

The VTK file format . **There are primarily two types of VTK files, legacy and XML**. We are going to focus on the legacy files, since they are a little bit ... Read more

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https://en.wikipedia.org › wiki › VTK

VTK

The Visualization Toolkit (**VTK**) is a free software system for 3D computer graphics, image processing and scientific visualization. Read more

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